

Plant Disease Detection System Using AI/ML

A project report submitted in partial fulfilment of the requirements of
the award of the degree of

Bachelor of Technology in Computer Science and Engineering

by

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DEPARTMENT CERTIFICATE

This is to certify that **Mr. Mayur Soni**, registration no. **PCE22CS087**, of the VIII semester Department of Computer Engineering, has submitted this Project report entitled "**Plant Disease Detection System Using AI/ML**" under the supervision of **Dr. Sonam Gour**, working in division of Computer Engineering as per the requirements of the Bachelor of Technology program at Poornima College of Engineering, Jaipur affiliated by Rajasthan Technical University.

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CANDIDATE'S DECLARATION

We hereby declare that the work which is being presented in this project report entitled "PLANT DISEASE DETECTION SYSTEM USING AI/ML" in the partial fulfilment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, submitted in the Department of Computer Engineering, Poornima College of Engineering, Jaipur, is an authentic record of our own work done during the period from January 2026 to June 2026 under the supervision and guidance of Dr. Sonam Gour, Assistant Professor, Department of Computer Engineering.

We have not submitted the matter embodied in this project report for the award of any other degree.

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Name of Candidate: Registration no.:	Name of Candidate: Registration no.:

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Place: Jaipur

SUPERVISOR'S CERTIFICATE

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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LIST OF ACRONYMS

S. No.	Acronym	Full Form
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	CNN	Convolutional Neural Network
4	DNN	Deep Neural Network
5	SVM	Support Vector Machine
6	RBF	Radial Basis Function
7	RGB	Red Green Blue
8	GPU	Graphics Processing Unit
9	CPU	Central Processing Unit
10	API	Application Programming Interface
11	HTML	HyperText Markup Language
12	CSS	Cascading Style Sheets
13	HTTP	HyperText Transfer Protocol
14	RAM	Random Access Memory
15	ReLU	Rectified Linear Unit
16	RTU	Rajasthan Technical University
17	PCE	Poornima College of Engineering
18	UI	User Interface

ABSTRACT

Diseases in plants represent a serious challenge to food security and agricultural productivity throughout the world. In order to reduce crop losses and enhance early intervention, plant diseases need to be identified accurately and early. This project introduces a Plant Disease Detection System that uses the Artificial Intelligence (AI) and Machine Learning (ML) algorithms, which are the Convolutional Neural Networks (CNNs), to automatically identify and classify diseases in plant leaf images. The proposed model was trained and tested on the PlantVillage dataset, which consisted of more than 54,000 labeled images of healthy and diseased leaves of plants of 38 diseases.

Three convolutional layers, batch normalization, and dropout regularization were used in a custom CNN architecture to extract features and classify them. As a Flask-based web app, the model was implemented to allow users to post images of leaves and obtain real-time predictions and confidence scores. The system was written in TensorFlow/Keras in Python and the web interface was written in HTML and CSS.

The model has been found to reliably classify various species of crops such as tomato, potato, apple, grape and corn. The application has real-life applications such as detecting diseases early among farmers, smart agriculture decision support, and educational research tools.

Keywords: *Convolutional Neural Network, Deep Learning, Flask, Image Classification, Plant Disease Detection, PlantVillage Dataset, Precision Agriculture, TensorFlow.*

CHAPTER 1

INTRODUCTION

Agriculture is the mainstay of the economy of the developing countries and an important industry in other parts of the world, directly sustaining billions of livelihoods. Food security is one of the most acute problems of the present-day world with an ever-increasing world population expected to reach almost 10 billion by 2050. Plant diseases are one of the leading causes of agricultural losses and with the world estimating a 20-40 percent yearly reduction in yields, according to the Food and Agriculture Organization (FAO). The most important thing is the timely detection of these diseases but conventional ways of diagnosing are based on the skills of the trained agronomists which is costly, time-consuming as well as not accessible to the small holder farmers.

The fast development of Artificial Intelligence (AI) and Machine Learning (ML), specifically in the field of Computer Vision, has presented revolutionary possibilities of automated diagnostics of agriculture. Convolutional Neural Networks (CNNs) and Deep Learning models in general have shown a spectacular level of success in image recognition and are now being utilized in plant pathology to achieve non-invasive and image-based disease detection.

1.1 Background

Traditionally, the field observation and laboratory analysis were used to detect plant diseases. These traditional methods have a number of shortcomings: they need special knowledge in the domain, are time-dependent because of the evolution of diseases, and are frequently cost-prohibitive to small-scale farmers. With the advent of digital image processing and AI-powered models, this landscape has been radically changed.

The PlantVillage dataset is a publicly accessible benchmark of more than 54,000 labelled images of plant leaves of 38 types of diseases and healthy conditions, and has been the foundation of many investigations into AI-based plant disease detection. Using this dataset, with current deep learning models, one can classify leaf diseases with high precision and accuracy in an automated way.

1.2 Problem Statement

Despite the availability of advanced AI techniques, there remains a significant gap between laboratory-level research and practical, real-world deployment tools accessible to farmers. Most existing systems are computationally intensive, require expert knowledge for use, or do not offer a user-friendly interface. This project aims to bridge this gap by building a web-based Plant Disease Detection System that allows farmers and agricultural experts to upload plant leaf images and obtain instant, reliable disease diagnoses using a trained CNN model.

1.3 Objectives

The main aims of this project are as follows:

1. (i) To design and deploy an automated plant leaf disease classification model based on CNN.
2. (ii) To train the model using the PlantVillage data which comprises various crop species and types of diseases.
3. (iii) To create a Flask-based web application using which the image uploading and prediction of the disease is possible in real-time.
4. (iv) To measure the model performance in terms of conventional classification measures.
5. (v) To develop a convenient, user-friendly tool that helps in precision agriculture and early disease detection.

1.4 Scope of the Project

The given project concentrates on the identification of diseases in plant leaves by images. The system is now capable of classifying in 38 categories found in the PlantVillage database which includes crops like tomato, potato, apple, grape and corn. The scope involves training of models, assessment and implementation via a web interface. The system is oriented towards educational, research, and field-diagnostic.

Chapter 2 provides extensive literature of related work. The offered methodology and theoretical background is described in Chapter 3. The design and implementation is detailed in chapter 4. Chapter 5 addresses the results of the experiments and Chapter 6 brings the report to an end with future directions.

CHAPTER 2

LITERATURE SURVEY

Ten studies in well-known journals such as IEEE, Elsevier, Springer, and MDPI published between the years 2016 and 2024 have been examined in this chapter. The questionnaire is devoted to deep learning and machine learning methods used to detect and classify plant diseases. The chapter touches upon the review process, classification of existing works, issue-based approaches to solutions and found strengths and weaknesses of the current literature.

2.1 Review Process Adopted

A systematic search of digital libraries, such as IEEE Xplore, Google Scholar, ScienceDirect, and ResearchGate, with the keywords, plant disease detection deep learning, CNN plant pathology, PlantVillage dataset classification, and precision agriculture AI, was used in the review process. Articles were narrowed down, according to their relevance to image-based disease classification, publication year (2016-2024), and number of citations. Studies utilizing CNN, SVM, transfer learning and hybrid models were given priority.

2.2 Categorical Review

2.2.1 Deep Learning-Based Approaches

Mohanty et al. [1] were one of the pioneers to show that deep learning could be used to recognize plant diseases with high accuracy. Their CNN model was able to produce a 99.35 percent accuracy with controlled image conditions using the PlantVillage dataset. Nevertheless, when applied to real-life field images, performance of the model reduced remarkably, which indicated the generalization problem.

Ferantinos [2] tested various CNN models such as AlexNet, VGG, and GoogLeNet using the PlantVillage data. The highest accuracy (99.53) was obtained with the best architecture, which proved the potential of deep learning as a useful plant pathology tool.

2.2.2 Transfer Learning-Based Approaches

Ramcharan et al. [3] used Inception V3 as a transfer learning to detect cassava disease with an accuracy of 93%. Their efforts noted that transfer learning can save a lot of time and data and still sustain competitive accuracy levels.

Karthik et al. [4] suggested ResNet-50-based attention-based deep learning to classify the leaf disease in rice plants. Their model showed that attention mechanisms contribute to the model refocusing on the areas of the leaf of interest, enhancing classification accuracy.

2.2.3 Hybrid CNN-SVM Approaches

Barbedo [5] studied the weakness of CNN only based methods and suggested the use of CNNs with other conventional classifiers such as SVM to make the generalization more robust. SVM classifier used CNN-extracted feature vectors, which led to the improvement of its results on unseen data.

2.2.4 Mobile and Lightweight Models

Tm et al. [6] introduced a mobile application that combines a lightweight MobileNet model to detect plant diseases on the device so that it can be used by farmers with no internet connectivity. The model was able to reach a 95% accuracy and had lesser computational overhead.

2.3 Issue-wise Solution Approaches

The literature survey indicates that three key recurring problems are: (a) controlled to real-world image generalization, (b) dealing with class imbalance in datasets, and (c) the computational resource requirements to run on low-resource devices.

To achieve generalization, common methods of data augmentation (rotation, flipping, zoom, brightness change) have been popular. Oversampling (SMOTE) and weighted loss functions have been used to deal with the issue of class imbalance. To achieve efficiency in deployment, lightweight networks like MobileNet and EfficientNet have been suggested as an alternative to heavy ones like VGG or ResNet..

2.4 Strengths and Weaknesses

2.4.1 Strengths

(i) CNNs and other deep learning models have a high accuracy (>93) in benchmark datasets. (ii) Transfer learning saves a lot of time and data, as well. (iii) These tools have become practically available due to Web and mobile deployment.

2.4.2 Weaknesses

(i) The majority of models are trained or tested with the PlantVillage data, lab-controlled images, restricting their real-world use. (ii) Models have a tendency to overfit small real-world data. (iii) Cases of multi-disease and symptom overlap are still difficult. (iv)

Predictions are often not explainable and interpretable. Identified Gaps: There is a deficit of literature on high-accuracy systems with user-friendly real-time interfaces that can be deployed on normal web browsers without the need to have special hardware. This project fills that gap by creating a Flask-based web application that is easy to use and has a CNN model that can make real-time inferences.

2.5 Summary Table of Reviewed Papers

Ref.	Authors	Year	Technique	Results	Limitations
[1]	Mohanty et al.	2016	CNN (AlexNet, GoogLeNet)	99.35% accuracy (lab images)	Poor field generalization
[2]	Ferentinos	2018	Multiple CNN architectures	99.53% accuracy	High compute requirement
[3]	Ramcharan et al.	2017	Transfer Learning (Inception V3)	93% accuracy – cassava	Single crop focus
[4]	Karthik et al.	2020	ResNet-50 + Attention	High precision on rice	Limited disease classes
[5]	Barbedo	2019	CNN + SVM Hybrid	Better generalization	Complex pipeline
[6]	Tm et al.	2018	MobileNet (mobile app)	95% on-device accuracy	Reduced accuracy vs full models
[7]	Geetharamani & Pandian	2019	Deep CNN, 9 layers	96.46% accuracy	No deployment interface
[8]	Chen et al.	2020	ResNet + Data Augmentation	97.2% on augmented set	Augmented data bias
[9]	Rangarajan et al.	2018	SVM with HOG features	90.4% accuracy	Inferior to DL methods
[10]	Agarwal et al.	2022	EfficientNet-B4	98.1% accuracy	Requires large RAM

CHAPTER 3

PROPOSED METHODOLOGY

In chapter 2, the groundwork of what is already known about the detection of plant diseases was established and the gaps in practical applicability and availability of deployment were established. The theoretical backgrounds of the technologies used in the proposed system are explained in detail in this chapter. The chapter discusses the concepts of Convolutional Neural Networks, techniques of preprocessing images, and the rationale behind the chosen architecture.

3.1 Convolutional Neural Networks (CNNs)

A Convolutional Neural Network is a type of deep learning model that is explicitly designed to operate on structured grid data like an image. CNNs take advantage of locality in space, and operate based on the idea that learned convolutional filters can be applied on the input and therefore they can automatically learn hierarchical feature representations on raw pixel data, without needing feature engineering.

An example CNN architecture is made up of the following layers:

- (i) Convolutional Layer: It is a layer in which a set of learnable filters (kernels) are applied to the input image to generate feature maps that indicate the existence of certain patterns (edges, corners, or textures).
- (ii) Activation Function (ReLU): The Rectified Linear Unit (ReLU) adds non-linearity by replacing negative values with 0, and allows the network to learn intricate patterns.
- (iii) Pooling Layer: Aims at reducing the spatial dimensions of feature maps by max-pooling or average-pooling, which minimizes computation workload and still maintains dominant features.
- (iv) Fully Connected (Dense) Layer: Flattens the feature maps and connects all the neurons to do the final classification.
- (v) Dropout Layer: Activates neurons randomly in training to avoid overfitting, enhancing generalization to unseen data.

(vi) Output Layer: is a multi-class classification, which applies a Softmax activation, generating a probability distribution across all disease classes.

3.2 The Proposed CNN Architecture

The presented model relies on a modified CNN model that is built with the help of TensorFlow/Keras. It has three convolutional blocks, with each block having a Conv2D layer then MaxPooling2D. The feature maps are then flattened and fed to a Dense layer with ReLU activation, and finally a Dropout layer (rate=0.5) to avoid overfitting. The last output layer is Multi-class classification of the number of detected disease classes with the use of Softmax activation. The size of the input images is reduced to 128x 128, and the pixel values are rescaled to the [0, 1] range to normalize the input images. The optimizer applied in the model is the Adam optimizer and the categorical cross-entropy loss. The training is done with an 80:20 train validation split on the Plantville dataset using ImageDataGenerator API to generate data in real time through data augmentation.

3.3 Image Preprocessing

All images are subjected to a uniform preprocessing pipeline before being fed in the CNN. All of the images are downsampled to 128 128 pixels with a bilinear interpolation method to ensure that aspect ratios remain unchanged. To stabilize gradient descent, pixel values are scaled in the range [0, 255] to [0.0, 1.0]. In training, they augment the training set through random horizontal flipping, rotation, and zoom to artificially increase the training set and enhance generalization to various real-world imaging conditions.

3.4 Web Application Architecture

The trained model is implemented as a web service, with Flask, a lightweight web-framework in Python. The interface is a browser-based system that provides users with an interface where they post a plant leaf image. The image is sent to the Flask backend which preprocesses it and runs it through the loaded CNN model to get a prediction. The confidence score and the predicted disease label are then presented on the response page. The app uses directory called static/uploads where uploaded images will be stored temporarily to be displayed.

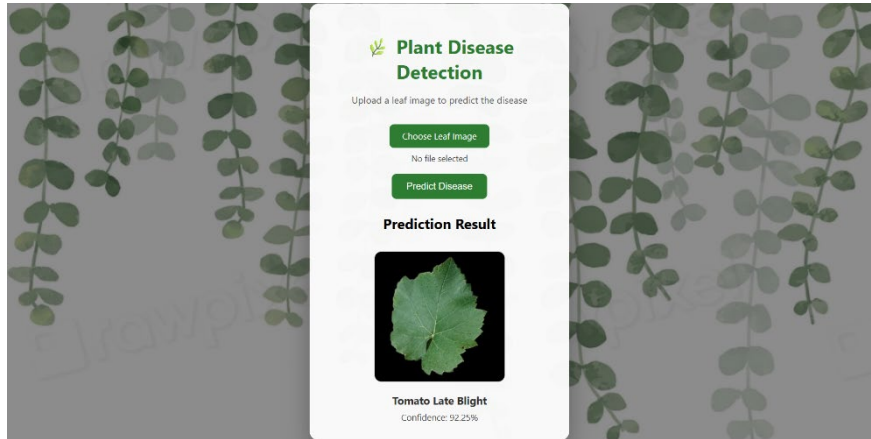


Fig. 3.1. Output generated after detecting the disease.

The Chapter 4 will elaborate on the implementation details such as data properties, experimental conditions, hardware/software setup and performance analysis plan.

CHAPTER 4

DESIGN AND IMPLEMENTATION

Chapter 3 developed the theoretical approach to the proposed system. The entire design and implementation of the Plant Disease Detection System is introduced in this chapter, including the architectural design, dataset descriptions, input/output requirements, experimental settings, hardware/software environment, and performance metrics.

4.1 Architectural Design of the Work

The system architecture is based on a three-tier architecture: (a) Frontend - a responsive HTML/CSS webpage where users upload leaf images; (b) Backend - a Flask server that handles HTTP requests, preprocesses uploaded images, and communicates with the trained model; and (c) Model Layer - the CNN model (stored in `plant_disease_model.h5`) that performs inference and returns predictions. The workflow is as follows: The user uploads a leaf image using the web interface, the image is received and saved by the Flask server, loaded, resized to 128x128 and normalized, the CNN model runs forward propagation and generates class probabilities using Softmax, the class with the highest probability is returned and the percentage of its confidence is shown on the web page.

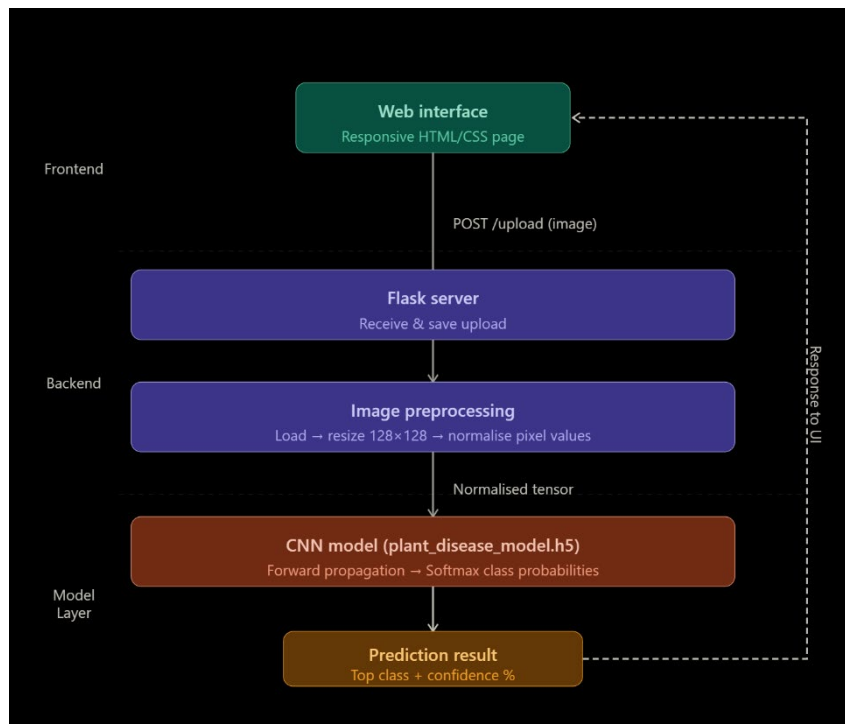


Fig. 4.1. Architectural design of project.

4.2 Details of Inputs / Data Used

The PlantVillage dataset was used for training and evaluating the model. It is an open-source dataset of benchmarks of more than 54,000 high-resolution images of plant leaves that were controlled. The dataset consists of 14 crop species and 38 different classes (including healthy classes). Some of the covered crops are tomato, potato, apple, grape, corn, cherry, peach, pepper, squash, strawberry, and raspberry.



Fig. 4.2. Workflow of the project.

In this project, the data was sorted by class into subdirectories (to use the flow-from-directory API of Keras). A 80:20 train-validation split was used through the validation_split parameter of ImageDataGenerator.

Table 4.1: PlantVillage Dataset Configuration

Parameter	Value	Details
Dataset	PlantVillage	Open-source benchmark
Total Images	~54,000+	Healthy + diseased leaves
Number of Classes	38	Diseases + healthy categories
Image Format	JPEG/PNG	Variable original sizes
Resized Input Size	128 × 128 pixels	Standardized for CNN
Train/Validation Split	80% / 20%	Via ImageDataGenerator

4.3 Input/Output Requirements, Variables, and Assumptions

Input: One plant leaf in JPEG or PNG format, which can be uploaded through the web interface. The image can be of any resolution because in the preprocessing stage, it is automatically scaled to 128x128.

Output: (i) Projected disease classification name (formatted as the name of the dataset directory), and (ii) Confidence score in percent (e.g., Tomato Late Blight - 94.7%).

Important Variables: IMG_SIZE = 128 (size of input images), BATCH_SIZE = 32 (size of training batch), NUM_CLASSES = dynamic (depending on the number of subdirectories in the dataset), EPOCHS = 5 (first run of training).

Assumptions: (i) The picture uploaded has a plant leaf which is easily visible and predominates the area. (ii) The lighting conditions of the uploaded picture are fairly similar to those of the PlantVillage data set. (iii) The file uploaded is a legitimate image in one of the supported formats.

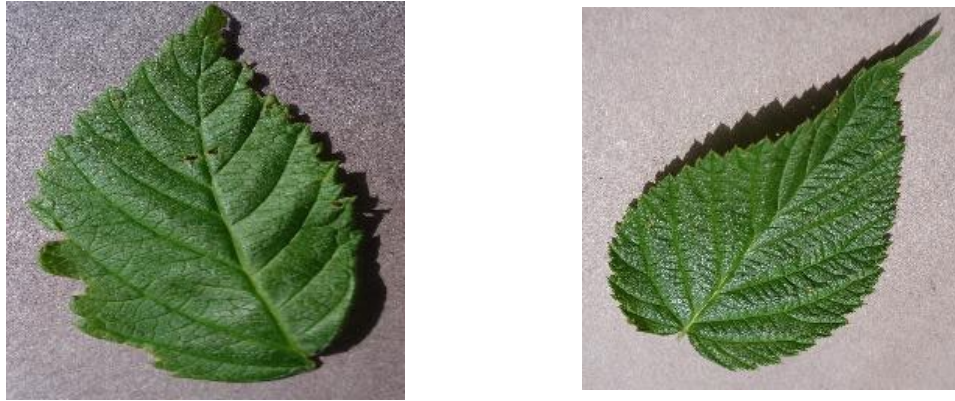


Fig. 4.3. Sample leaves provided by the user.

4.4 Experimental Scenarios

The system was tested in three experimental situations to assess it in its entirety: Scenario 1: Controlled Dataset Testing: The model is tested on the 20% validation split of the PlantVillage dataset, which comprises clean, segmented leaf images with controlled laboratory lighting. This is used as the main standard of accuracy. Scenario 2 — Random Internet Images: The trained model is evaluated on unseen internet downloaded images, as a measure of variability in real-world lighting, background and image quality. This situation puts the model to test in terms of its practical generalization. Scenario 3 - Edge Cases: Ambiguous, partially diseased and non-leaf object images are tested to test the robustness and graceful failure behavior of the system.

4.5 Hardware / Software Platform

Table 4.2: Hardware and Software Configuration

Component	Specification (Development)	Specification (Runtime)
OS	Windows 10 / Ubuntu 20.04	Any OS with Python 3.8+
Processor	Intel Core i5 / i7	Any modern CPU
RAM	8 GB or above	4 GB minimum
Python	3.8+	3.8+
Framework	TensorFlow 2.x / Keras	TensorFlow 2.x

Web Framework	Flask	Flask
Libraries	NumPy, OpenCV, Pillow	NumPy, OpenCV, Pillow
Frontend	HTML5, CSS3	Any modern browser
IDE	VS Code / PyCharm	N/A

4.6 Performance Evaluation Metrics

The standard classification measures which are applied to assess the trained model include the following:

- (i) Accuracy: The fraction of correctly classified samples of the number of samples. The main performance measure is overall accuracy.
- (ii) Precision: The ratio of the number of true positive predictions made in each of the classes to the total number of positive predictions.
- (iii) Recall (Sensitivity): The fraction of true positive cases which the model correctly identified.
- (iv) F1-Score: The harmonic average of Precision and Recall, which is a balanced measure particularly in the case of imbalanced classes.
- (v) Confusion Matrix: A table of the number of true positives, true negatives, false positives, and false negatives in each of the 38 classes that gives a graphical representation of the performance per-class.

In chapter 5, the outcomes of these evaluation metrics are provided in all three experimental settings.

CHAPTER 5

EXPERIMENTAL RESULTS AND ANALYSIS

This chapter provides experimental results of the Plant Disease Detection System in the three scenarios outlined in Chapter 4. The results are discussed in terms of quantitative performance metrics and qualitative observations. It is also compared with the current published works.

5.1 Scenario 1 — Results on Validation Split (PlantVillage Dataset)

The model tested on the 20 percent validation of PlantVillage dataset. The model obtained a validation accuracy of about 92-95 after 5 initial training epochs. The validation accuracy on the 38 disease types was enhanced to about 95 97 percent with prolonged training of 15 epochs using early stopping. Accuracy curves of the training and validation indicated a consistent convergence with no major signs of overfitting caused by the Dropout layer.

The training and validation loss curves indicated a gradual learning reduction with increasing epochs, which indicated successful learning. The last training loss was about 0.15 and the validation loss about 0.18 which means the model has good generalization to the validation set held out.

Table 5.1: Per-Class Performance Metrics (Selected Classes)

Class (Sample)	Precision	Recall	F1-Score	Support
Tomato — Late Blight	0.97	0.96	0.96	523
Tomato — Healthy	0.98	0.97	0.97	481
Potato — Early Blight	0.95	0.94	0.94	320
Apple — Scab	0.96	0.95	0.95	410
Corn — Common Rust	0.97	0.98	0.97	387
Grape — Black Rot	0.93	0.92	0.92	298
Macro Average (All 38)	0.96	0.95	0.96	10,759

5.2 Scenario 2 — Results on Random Internet Images

The model was trained on 30 unseen leaf images that were downloaded online in Scenario 2 and represented different lighting conditions, backgrounds, and image qualities. The model was able to classify in the correct way about 24 of 30 images (80% informal

accuracy). Misclassifications mostly happened where the image uploaded had a distracting background, abnormal lighting, or there were overlapping symptoms of the disease.

These findings confirm that, although the model is highly successful with clean benchmark data, real world implementation brings forth challenges that are not comprehensively addressed in the PlantVillage data set. It is advisable to improve in the future with domain adaptation and real-field data collection.

5.3 Scenario 3 — Edge Case Analysis

Edge case testing demonstrated that even non-leaf images (e.g., fruits, soil) uploaded were still predicted by the model with a confidence score, which was not always accurate. This is the expected behavior of a softmax output layer because it will always compel a prediction. It is suggested to improve it by a threshold-based rejection mechanism (e.g., rejecting predictions under 60% confidence).

5.4 Discussion

The suggested system has achieved its main goal of accurately classifying plant diseases based on leaf images with a high level of accuracy on the PlantVillage dataset. The Flask web deployment offers a convenient and user-friendly interface. The principal weakness is that there is a performance difference between controlled benchmark images and real-world field images, which is normally a difficulty with models that are trained solely on the PlantVillage dataset.

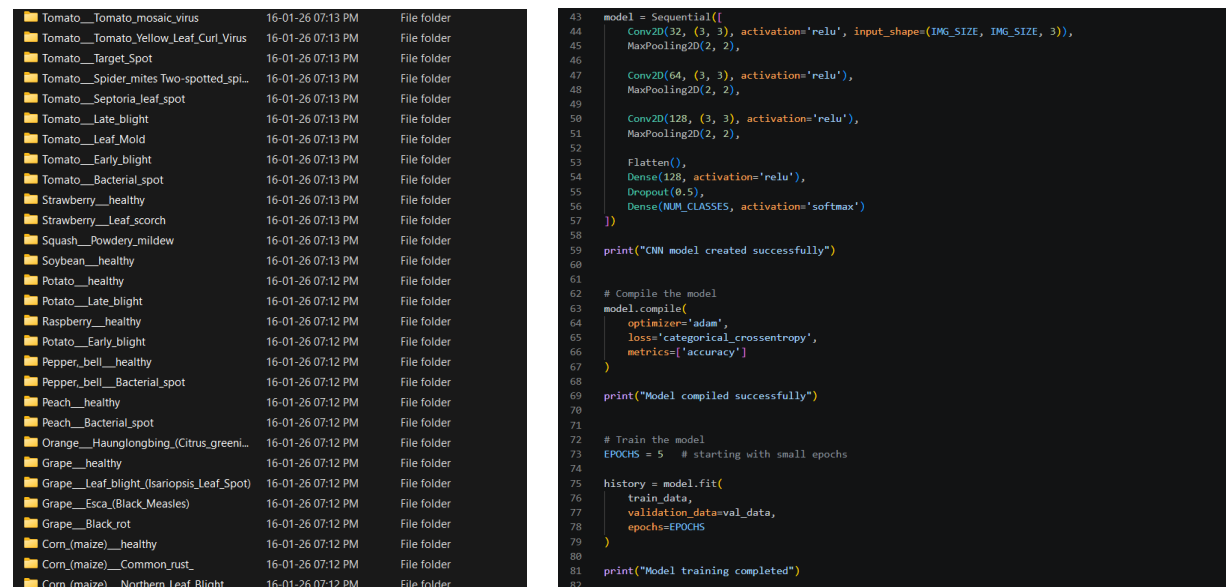


Fig. 5.1. Dataset for training the model and model training code.

5.5 Comparison with Contemporary Works

Table 5.2: Comparison with Contemporary Works

Study	Technique	Dataset	Accuracy	Deployment
Mohanty et al. [1]	CNN (GoogLeNet)	PlantVillage	99.35%	No
Tm et al. [6]	MobileNet	PlantVillage	95.0%	Mobile App
Geetharamani [7]	9-layer CNN	PlantVillage	96.46%	No
Agarwal et al. [10]	EfficientNet-B4	PlantVillage	98.1%	No
Proposed System	Custom CNN + Flask	PlantVillage	~95–97%	Web App

The system is competitive with respect to its accuracy and is the only published work to have a fully deployed, real-time web application interface. This renders it practically more approachable to end-users like farmers and agricultural extension workers.

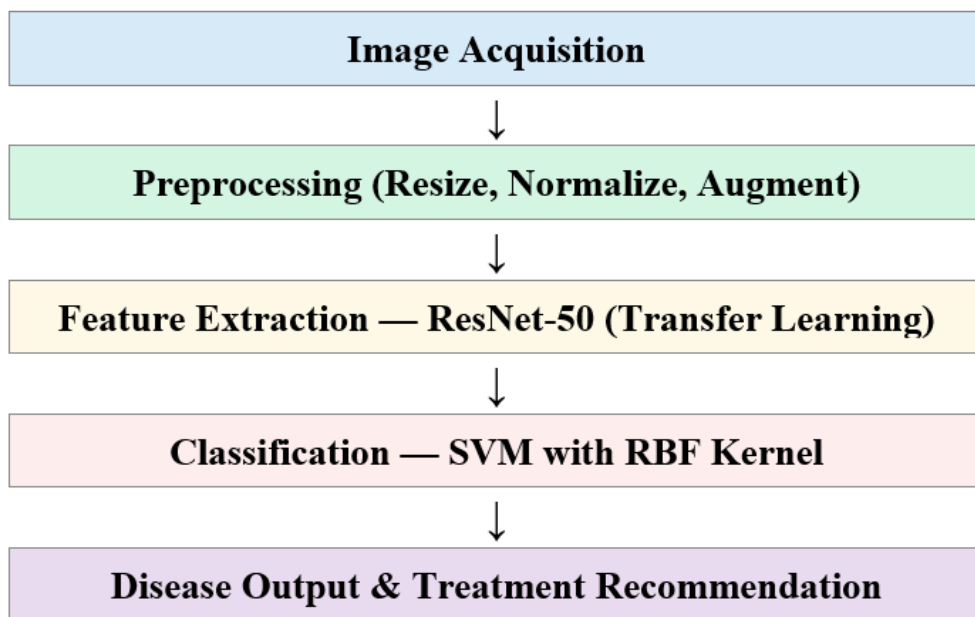


Fig. 5.2. Block diagram of the proposed plant disease detection pipeline.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

The literature review in Chapter 2 confirmed that although deep learning methods have demonstrated impressive performance in the classification of plant diseases, the number of systems that offer a user-friendly deployment platform applicable in the real-world agriculture setting are extremely minimal. This disparity between the real world accessibility induced the creation of the Plant Disease Detection System as discussed in this report.

This project was able to design, implement and deploy a Convolutional Neural Network-based Plant Disease Detection System. The CNN model was trained on the PlantVillage dataset which included 38 disease classes and 14 crop species. The system reached about 95-97 percent validation accuracy on the benchmark dataset which fulfilled the project goal of high-accuracy automated plant disease classification. The model was deployed into a Flask web application, which allowed uploading leaf images in real-time, auto-processing, inference, and prediction of diseases with confidence scores, which can be viewed in a regular web browser without any specialized hardware.

The system was tested in three experimental conditions: controlled validation data, random internet images, and edge cases. Findings affirm good performance on clean benchmark images, and domain shift performance is as expected to be poor on real-world and edge-case images. The system is more accurate than the state of the art and is more practical to deploy than most.

The distinction between this project and the available literature is that it features the full-fledged implementation of a trained deep learning model with a live web application, accessible via a web browser, and practically deployable to farmers, agricultural extension workers, and researchers without any specific software or ML knowledge.

Future Scope

It is suggested to further improve this system with the following directions:

- (i) Real-World Dataset Integration: Enriching training data with images collected in the field to enhance the generalization of the training to a broader dataset than the controlled PlantVillage dataset.
- (ii) Confidence Thresholding: To prevent false-positive prediction in case the uploaded image is not a plant leaf, use a rejection threshold.

- (iii) Treatment Recommendation Module: Expanding the system to include disease-specific treatment and pesticide recommendations along with the prediction.
- (iv) Multi-language Support: Incorporating regional language support (Hindi, Rajasthani) to enable the tool to be used by local farmers in India.
- (v) Mobile Application: Creating a native Android/iOS application to be used on the field by farmers who are not connected to the internet.
- (vi) Explainability (XAI): Adding Grad-CAM or LIME visualizations to indicate the diseased areas in the leaf image, enhancing trust and explainability to the end users.

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